

1. Administrative & Study Identification

Full Study Title: A Randomized, Phase III Study of Rilvegostomig in Combination With Fluoropyrimidine and Trastuzumab Deruxtecan Versus Trastuzumab, Chemotherapy, and Pembrolizumab for the First Line Treatment of HER2-positive Gastric Cancer (ARTEMIDE-Gastric01) **Short Title/Acronym:** ARTEMIDE-Gastric01 **Protocol Number:** D702AC00001 **NCT Number:** NCT06764875 **Sponsor Name:** AstraZeneca **Investigational Product:** Rilvegostomig, Trastuzumab deruxtecan (T-DXd), 5-fluorouracil, Capecitabine, Pembrolizumab, Trastuzumab, Cisplatin, Oxaliplatin **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of rilvegostomig in combination with fluoropyrimidine and T-DXd (Arm A) compared to the standard of care—trastuzumab, chemotherapy, and pembrolizumab (Arm B)—as a first-line treatment in HER2-positive locally advanced or metastatic gastric or gastroesophageal junction (GEJ) adenocarcinoma participants whose tumors express PD-L1 CPS ≥ 1 . **Secondary Objectives:** To evaluate the contribution of each component in the experimental arm by comparing it to a separate arm (Arm C) receiving rilvegostomig in combination with trastuzumab and chemotherapy.

2.2 Background & Rationale Gastric cancer is often diagnosed at an advanced stage with a poor prognosis. HER2 overexpression occurs in roughly 20% of cases. While adding immune checkpoint inhibition to trastuzumab and chemotherapy has shown clinical benefit, there is an unmet need for improved targeted therapies. Rilvegostomig is a bispecific, humanized monoclonal antibody targeting both PD-1 and TIGIT receptors. Trastuzumab deruxtecan (T-DXd) is an antibody-drug conjugate targeting HER2. This study investigates whether the dual checkpoint blockade of rilvegostomig combined with the targeted cytotoxicity of T-DXd and fluoropyrimidine improves clinical outcomes compared to current standard regimens.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Trastuzumab, chemotherapy, and pembrolizumab) **Blinding:** Open-label, Sponsor-blinded **Randomization:** Randomized (3-arm)

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 840 participants **Number of Sites:** Up to 200–250 sites globally across approximately 25 countries **Estimated Study Start:** Currently Active/Recruiting **Estimated Primary Completion:** Total study participation expected to last about 2 years

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age. 2. Previously untreated, unresectable, locally advanced or metastatic gastric or GEJ adenocarcinoma. 3. HER2-positive for gastric cancer on a recent tumor biopsy. 4. PD-L1 combined positive score (CPS) ≥ 1 . 5. Provision of tumor tissue sample from a recent biopsy adequate for HER2 and PD-L1 testing. 6. Measurable target disease assessed by the Investigator based on RECIST v1.1. 7. WHO or Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 8. Adequate organ and bone marrow function within 14 days before randomization; LVEF $\geq 55\%$ within 28 days before randomization. 9. Adequate treatment washout period before randomization.

4.2 Exclusion Criteria 1. Spinal cord compression or brain metastases, unless asymptomatic, treated, stable, and not requiring corticosteroids or anticonvulsants (and recovered from acute toxic effects of radiotherapy). 2. Uncontrolled infection including tuberculosis and active hepatitis A, or uncontrolled infection requiring intravenous (IV) antibiotics, anti-virals, or antifungals. 3. History of (non-infectious) interstitial lung disease (ILD)/pneumonitis, current ILD/pneumonitis, suspected ILD/pneumonitis that cannot be ruled out by imaging at screening, or lung-specific intercurrent clinically significant illnesses. 4. Active or prior documented autoimmune or inflammatory disorders requiring chronic treatment with steroids or other immunosuppressive treatment. 5. History of another primary malignancy, except for malignancy treated with curative intent with no known active disease within 3 years before the first dose of study intervention and of low potential risk for recurrence. 6. Lack of physiological integrity of the upper gastrointestinal tract. 7. Known dihydropyrimidine dehydrogenase (DPD) enzyme deficiency. 8. Contraindication to pembrolizumab or trastuzumab, or contraindications to fluoropyrimidine (5-FU and capecitabine) or platinum (cisplatin and oxaliplatin) treatment as per local label. 9. Persistent toxicities caused by previous anti-cancer therapy. 10. Recent receipt of a live, attenuated vaccine. 11. Chronic/active HBV or HCV infection unless controlled. 12. Clinically significant cardiac or psychological conditions. 13. Any active non-infectious skin disease requiring systemic treatment or history of drug-induced severe cutaneous adverse reaction. 14. A pleural effusion, ascites, or pericardial effusion that requires drainage, peritoneal shunt, or cell-free and concentrated ascites reinfusion therapy (CART). 15. Any concurrent anti-cancer treatment with the exception of receptor

activator of nuclear factor kappa-B ligand inhibitors. 16. Have had a major surgical procedure recently (excluding placement of vascular access), recent significant traumatic injury, or an anticipated need for major surgery during the study. 17. Current or prior use of immunosuppressive medication within 14 days before study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency & Route of Administration:

Administered in 3-week cycles. * **Arm A:** Intravenous (IV) rilvegostomig + IV T-DXd on Day 1 of each cycle + Investigator's choice of capecitabine (oral, twice daily for 14 days) or 5-fluorouracil (IV continuous infusion over 24 hours for the first 5 days). * **Arm B:** IV pembrolizumab + IV trastuzumab + Investigator's choice of FP (5-FU and cisplatin) or CAPOX (capecitabine and oxaliplatin). * **Arm C:** IV rilvegostomig + IV trastuzumab + Investigator's choice of FP or CAPOX. **Duration of Treatment:** Until unacceptable toxicity or progressive disease (as defined by RECIST v1.1). Patients who continue to show clinical benefit can stay on treatment post-progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced side effects (e.g., antiemetics). **Prohibited:** Current or prior use of immunosuppressive medication within 14 days before study intervention; concurrent anticancer treatment (with the exception of RANK ligand inhibitors).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (HER2/PD-L1), LVEF, Blood Tests
Baseline	Day 0	Randomization, First Dose
Interim	Every 3 Weeks	Treatment administration, Safety Labs, Efficacy Assessment (CT scans), Questionnaires
End of Study	Follow-up	End of treatment evaluation, survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy and safety evaluations of antitumor activity. * **Progression-Free Survival (PFS):** Defined as the time from randomization until progression per RECIST v1.1, or death due to any cause. * **Overall Survival (OS):** Defined as the time from randomization until the date of death due to any cause.

7.2 Secondary & Exploratory Endpoints * Contribution of each component (Comparison of Arm C vs Arm A). * Objective Response

Rate (ORR). * Duration of Response (DoR). * Proportion of all randomized participants alive and progression-free at 6 months (PFS6). * Incidence and severity of adverse events (Safety and Tolerability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine blood tests, physical examinations, and patient-reported questionnaires at each treatment cycle. Specific monitoring for Interstitial Lung Disease (ILD) and pneumonitis due to T-DXd and immune-related AEs from rilvegostomig. **Serious Adverse Events (SAEs):** Typical 24-hour reporting protocols for expedited safety monitoring. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Approximately 840 patients randomized to ensure adequate power to detect statistically significant improvements in efficacy outcomes between the experimental combinations and the standard of care. **Handling of Missing Data:** Standard oncology imputation protocols for missing clinical or radiographic assessments.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol adherence, drug accountability, and data integrity. **Audit Trail:** Validated eCRF systems with comprehensive audit trails for data entry and queries.

11. Study Identifiers & Governance

Posting ID: 1059499775530863043 **Primary ID:** D702AC00001 **Registry Number:** NCT06764875 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMIDE-Gastric01 **Official Regulatory Title:** A Phase I Study of Rilvegostomig in Combination With Fluoropyrimidine and Trastuzumab Deruxtecan as the First-line Treatment for HER2-positive Gastric Cancer

12. Clinical Framework (Gastric Cancer Specific)

Primary Condition: Gastric or Gastroesophageal Junction (GEJ) Adenocarcinoma **Diagnosis Threshold:** HER2-positive and PD-L1 CPS ≥ 1 **Medical Methodology:** Targeted HER2-directed antibody-drug conjugate combined with dual immune checkpoint blockade (PD-1/TIGIT) and systemic chemotherapy. **Optimization Target:** Maximizing anti-tumor response and extending progression-free/overall survival in the first-line setting.

13. Study Procedures & Methodology

Management Pathway: Standard oncological clinical pathway for metastatic/advanced gastric cancer. **Education Protocol (HCP):** Site initiation visits and ongoing safety training, particularly regarding the identification and management of ILD/pneumonitis. **Education Protocol (Patient):** Informed consent process, toxicity reporting education, and self-management instructions for oral chemotherapy (capecitabine). **Quality Audit Mechanism:** Centralized radiological review and clinical data management audits.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity (approx. 2 years). **Onsite Visit Cadence:** Every 3 weeks for treatment cycles (Day 1 of each 21-day cycle). **Remote Monitoring Frequency:** Between cycles via telehealth/phone if necessary for symptom management. **Checklist Review Interval:** Prior to each infusion cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				